

Andy Rodriguez

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Summary

- Senior Software Engineer with over 10 years of hands-on experience in designing and developing highly scalable, high-performance systems.
- Specializing in **cloud-native development, microservices architecture, data engineering, AI/ML, React, Node.js, and DevOps** practices.
- Expert in building secure, maintainable, and efficient software solutions, leveraging technologies like **Python, Go, C#/.NET, Java, AWS, Kubernetes, and Docker**.
- Experienced in leading cross-functional teams, mentoring engineers, and driving technical initiatives that improve both system performance and business outcomes.
- Passionate about leveraging cutting-edge technologies to create innovative solutions in cloud, data, and AI/ML domains.

Education

- **UNIVERSITY OF MIAMI – MIAMI, FL**
- Bachelor of Science in Computer Science
- Sep 2011 – Jun 2015

Skills & Abilities

Programming Languages & Frameworks: Python (Django, Flask, FastAPI), Go (Golang, Gin, Echo), C#/.NET (ASP.NET Core, Entity Framework), Java (Spring Boot, Hibernate), React, Node.js, JavaScript, TypeScript, Ruby (Ruby on Rails, Sinatra)

Cloud Technologies & DevOps: AWS (EC2, Lambda, S3, RDS, DynamoDB, CloudFormation), Azure, Google Cloud Platform (GCP), Docker, Kubernetes, Terraform, Jenkins, GitLab CI/CD, CircleCI, Ansible, GitOps, Helm

Microservices & Distributed Systems: Microservices architecture, API Gateway, gRPC, Event-driven architecture, Service Mesh (Istio), RabbitMQ, Kafka, Redis, CQRS, GraphQL, WebSockets

Security & Authentication: OAuth2, JWT, SSO, MFA, SSL/TLS, Data Encryption, API Security, Penetration Testing, OWASP, IAM, Firewalls, Vulnerability Scanning

Database Technologies: SQL (PostgreSQL, MySQL, MS SQL Server), NoSQL (MongoDB, Cassandra, DynamoDB), Elasticsearch, Redis, Data Sharding, Replication, Database Optimization, ETL Pipelines, Data Warehousing

AI/ML & Data Engineering: TensorFlow, Keras, Scikit-Learn, Data Pipelines, Real-Time Analytics, MLOps, Natural Language Processing (NLP), Data Modeling, Machine Learning Integration

Testing & Quality Assurance: TDD, PyTest, RSpec, JUnit, Mocha, Selenium, Postman, Cypress, Load Testing, End-to-End Testing, Automated Testing

CI/CD & Automation: Jenkins, GitLab CI/CD, CircleCI, Docker Compose, Helm, Automated Testing, Deployment Automation, Cloud-native CI/CD Pipelines, Release Management

Monitoring & Observability: Prometheus, Grafana, ELK Stack, Datadog, CloudWatch, New Relic, OpenTelemetry, Distributed Tracing, Alerting

Leadership & Collaboration: Agile Scrum, Kanban, Technical Leadership, Mentorship, Code Reviews, Cross-functional Collaboration, Stakeholder Management, Product Management

Professional Experience

Senior Software Engineer | United Health Group – Miami, FL | Jan 2021 – Present

- Designed and developed **HIPAA-compliant systems** for managing healthcare data, leveraging **AWS Lambda, DynamoDB, S3, and EC2** for scalability and performance.
- Led the migration of a legacy monolithic application to a **microservices architecture** using **Go** and **Java (Spring Boot)**, increasing deployment efficiency by 40%.
- Implemented **real-time data processing** using **Kafka** and **Redis** for faster decision-making, improving system throughput by 50%.
- Developed **secure RESTful APIs** using **Go, Python (Flask)**, and **Java** to provide secure and efficient communication between services.
- Spearheaded the use of **Kubernetes** and **Docker** for container orchestration, enhancing deployment automation and system stability.
- Designed and implemented **CI/CD pipelines** using **GitLab CI/CD** and **Jenkins**, reducing deployment time by 35%.
- Optimized database performance by implementing **SQL query tuning, indexing, and sharding** with **PostgreSQL**, improving system speed by 30%.
- Led the integration of **machine learning models** into the platform using **TensorFlow** and **Scikit-Learn**, enhancing decision support and predictive analytics.
- Worked with stakeholders to gather requirements and ensure technical solutions align with business goals, increasing customer satisfaction by 20%.
- Implemented **multi-factor authentication (MFA)** and **OAuth2** for robust security, ensuring compliance with healthcare regulations.
- Mentored junior engineers in **cloud architecture, DevOps practices, and agile development** methodologies, boosting team productivity by 25%.
- Set up **real-time monitoring** using **Prometheus, Grafana, and Datadog**, improving incident response time by 50%.
- Led initiatives to implement **serverless architectures** using **AWS Lambda**, reducing infrastructure costs by 20%.

- Enhanced application security by integrating **JWT** for secure authentication and implementing **SSL/TLS** for secure communication.
- Designed and deployed **event-driven architecture** using **AWS SNS/SQS**, improving system responsiveness and fault tolerance.
- Contributed to open-source **cloud-native projects**, sharing best practices for building scalable, secure microservices architectures.
- Conducted **code reviews** and **unit testing** to ensure code quality and maintainability.
- Delivered data engineering solutions by setting up **ETL pipelines** and **data warehouses** using **AWS Redshift** for analytics.
- Implemented **data governance** and compliance strategies, ensuring secure and effective handling of sensitive healthcare data.

Software Engineer | Netflix – Los Gatos, CA | Jun 2018 – Dec 2020

- Developed **microservices** for Netflix's video streaming platform, ensuring **low-latency** delivery and **high availability** for millions of global users.
- Led the implementation of **Kubernetes clusters** across multiple AWS regions, improving system fault tolerance and resource scaling.
- Created real-time **video analytics** pipelines using **Apache Kafka** and **Apache Flink**, enabling enhanced monitoring and failure prediction.
- Optimized video content delivery using **CDN** and **WebSocket** for better real-time user experience.
- Integrated **OAuth2** for secure, token-based authentication across services, ensuring system-wide security.
- Collaborated with **data engineers** to deploy **machine learning models** that personalized content recommendations, increasing user engagement by 15%.
- Enhanced **API security** by adopting **JWT** and **SSO**, securing sensitive user data and improving platform integrity.
- Established a **CI/CD pipeline** for automated testing and deployment using **Jenkins** and **GitLab CI/CD**, reducing downtime by 30%.
- Worked on **distributed systems** to ensure scalability of video processing, reducing latency and increasing system throughput by 40%.
- Contributed to **open-source projects**, focusing on **cloud-native development** and **microservices patterns**.
- Implemented **serverless solutions** with **AWS Lambda**, improving scalability while reducing operational costs.
- Managed integration of **real-time error tracking** using **Datadog** and **Prometheus**, enhancing monitoring and reducing incident resolution time by 50%.

- Created **fault tolerance mechanisms** using **Kafka** to ensure continuous video streaming during outages.
- Set up and maintained **data pipelines** for **real-time analytics**, providing real-time insights into content quality and user experience.
- Led a team to migrate legacy systems to **cloud-native infrastructure**, improving system reliability and flexibility.
- Participated in **peer code reviews** and provided mentorship on best practices for **microservices design** and **cloud security**.

Software Designer | Kforce – Miami, FL | Jul 2015 – May 2017

- Developed and deployed enterprise-level applications using **C#/.NET** and **Java**, ensuring high performance and scalability for business-critical systems.
- Built and maintained **ETL pipelines** using **Apache Spark** for data transformation and storage in **AWS S3**.
- Developed backend **APIs** with **Node.js** and **Ruby on Rails** to ensure smooth integration with client systems and third-party services.
- Worked with the team to design and implement **cloud infrastructure** using **AWS EC2, S3, and RDS**, improving operational efficiency by 20%.
- Created automated deployment scripts with **Ansible** and **Docker**, reducing manual errors and deployment times by 40%.
- Integrated **real-time data streaming** solutions using **Apache Kafka** and **RabbitMQ**, increasing data flow efficiency.
- Led a project to **containerize** applications using **Docker**, improving environment consistency and deployment reliability.
- Improved database performance through **query optimization, indexing, and sharding** with **MySQL** and **PostgreSQL**.
- Collaborated with **cross-functional teams** to define technical specifications and ensure timely delivery of system features.
- Worked closely with **data engineers** to implement **data warehousing** and **analytics solutions** using **AWS Redshift**.
- Implemented **security best practices** in system design, including **encryption** and **firewall management**.
- Delivered key technical solutions for **frontend applications** using **React** and **Node.js**, improving user experience by 30%.
- Participated in **Agile Scrum** meetings, helping prioritize features and sprints based on business goals and technical requirements.

Key Projects

Healthcare Data Analytics Platform

Developed a **cloud-native platform** for processing and visualizing healthcare data using **AWS Lambda** and **Kinesis**, reducing data processing times by 50%. Implemented **machine learning models** for predictive analytics, improving decision-making processes.

Real – Time Video Processing System

Engineered a **highly scalable video processing platform** at Netflix using **Go**, **Kafka**, and **Kubernetes** for distributed processing. The system improved content delivery and user engagement by reducing buffering by 30%.

AI/ML – based Content Recommendation System

Integrated **TensorFlow** and **Scikit-Learn** into a real-time video recommendation engine, which led to a 15% increase in user retention on the Netflix platform.

Certifications

- **AWS Certified Solutions Architect – Associate**
- **Certified Kubernetes Administrator (CKA)**
- **Microsoft Certified: Azure Developer Associate**
- **Certified Scrum Master (CSM)**
- **AWS Certified DevOps Engineer – Professional**

Technical Blogs & Open Source Contributions

- Contributor to **open-source projects** focusing on **cloud-native technologies**, **microservices**, and **DevOps** practices.
- Regular speaker at **local tech meetups** and contributor to **Medium.com** articles on **scalable architectures**, **cloud technologies**, and **AI/ML development**.

Additional Information

- Fluent in English.
- Actively mentors junior engineers and helps foster a **collaborative and inclusive** work environment.
- Strong advocate for **work-life balance**, ensuring high team morale and sustainable long-term productivity.